TIRTHARAJ BHATTACHARYA



Electronics & Communication Engineering Student | GNIT, Kolkata

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Linkedin | Github | Portfolio | Resume

About Me

B.Tech (ECE) undergraduate with interests in Embedded Systems, Artificial Intelligence, IoT, and software development, with a focus on integrating intelligent behaviour into real-world systems. Worked on collaborative projects involving embedded systems, web applications, and exploratory AI/ML implementations, gaining practical exposure to machine learning workflows and embedded technologies. Interested in building context-aware and adaptive systems through the intersection of hardware, software, and AI.

Education

Guru Nanak Institute of Technology – B.Tech, ECE (2024–Present)

Central Model School, Barrackpore – AISSCE, PCMB (2021–2023): 90%

Central Model School, Barrackpore – AISSE (2008–2021): 94%

DAV Public School, Barrackpore – (2008–2019)

Experience

Robotics Trainee, Ardent Computech Pvt Ltd | Internship (July, 2026)

AIML Trainee, Ardent Computech Pvt Ltd | Internship (March, 2026 – April, 2026)

Content Writer, Euphonous X Zyzzva 2k26 | Volunteer (December, 2025 – March, 2026)

Content Writer, Euphonous X Zyzzva 2k25 | Volunteer (February, 2025 – September 2025)

Technical Skills

Programming Languages: Python, C, HTML, CSS

AI/ML & Libraries: TensorFlow, Keras, OpenCV, NumPy, Pandas

Hardware & Tools: Arduino Uno, ESP32, IoT Prototyping, Breadboarding

Core Competencies:

- Public Speaking (Debates & Speeches)
- Leadership
- Team Collaboration
- Content Writing
- Presentations

Research and Publications

Research Paper – Co-author

Design And Development Of A Resonance-Tuned Hybrid Piezoelectric-Electromagnetic Energy Harvesting System *(Published in IJCRT)*

• Contributed to manuscript editing, refinement, and review alongside co-authors.

• Participated in collaborative discussions and content preparation during the publication process.

Projects

NeuroKavach | Idea Manthan Project (2nd Position)
Technologies: Haptic Feedback Systems, Wearable Technology
• Proposed a wearable haptic feedback concept for real-time cognitive intervention
• Collaborated with teammates to refine functionality and presentation
• Secured 2nd position through concept development and solution presentation

Digital Fort Knox | Cryptonix-V1 Project
Technologies: ESP32, IoT, Device Monitoring
• Worked with a team to develop a security-focused embedded solution
• Contributed to solution design, feature discussions, and prototype development.

AI Study Assistant
Technologies: Python, Streamlit, Gemini API
• Built an AI study assistant using Python, Streamlit, and Gemini API for personalized learning.
• Implemented AI chat, quiz generation, study planning, and performance analysis.

IoT Fire Alert and Automatic Sprinkler System | ESP32-Based Fire Alert and Notification System
Technologies: ESP32, IoT, Blynk
• Integrated an IR flame sensor, relay-controlled water pump, LED, and buzzer for automated fire response.
• Enabled Wi-Fi connectivity with **Blynk IoT** for real-time smartphone notifications.

Emotion Detection Project | BCT Training Program
Technologies: Python, OpenCV
• Implemented an emotion detection system during technical training activities.
• Gained exposure to computer vision workflows and external library integration.
• Learned practical implementation alongside theoretical Python concepts.

MatsyAI | Hackforge Project
Technologies: Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Scikit-learn
• Worked collaboratively in a team to design and develop a proposed solution
• Participated in brainstorming, solution planning, and project development within a time-constrained environment

Plant Disease Detector
Technologies: Python, TensorFlow
• Built an experimental machine learning project for identifying plant diseases from image data.
• Explored image preprocessing and basic classification workflows.
• Used the project as a practical introduction to machine learning concepts.

Screen Recorder
Technologies: Python, Pyautogui
• Developed a utility for capturing on-screen activity using Python libraries.
• Explored system interaction and screen capture functionality.

Voice Recorder
Technologies: Python, NumPy, SoundDevice
• Developed a utility for recording audio input using Python audio-processing libraries.

• Explored concepts related to audio capture and handling.

Kitab-Al Hind (Book Selling Website)
Technologies: HTML, CSS
• Built a responsive website interface for displaying and organizing book information.

Achievements and Competitions

• Earned 2nd Position at *Idea Manthan* organized by IEM for **NeuroKavach**
• Earned 1st Position at *Eureka* organized by GNIT for **Gurdian Mat,** contributing to concept presentation.
• Participated in *Cryptonix-V1* organized by Heritage Institute of Technology with **Digital Fort Knox**
• Selected at Institute-level screening for ***Smart India Hackathon (SIH)***
• Participated in *Hackforge* organized by Jadavpur University with **MatsyAI**

Positions of Responsibility

Vice President – GNIT Literary Club | 2025 – Present

Content & Communication Activities

• **Content Creator — The Anime Maniacs (YouTube)**
Built a content channel reaching **250K+ cumulative views** and **400+ subscribers**

• **Freelance Blogger**
Published **10+ articles** across online platforms

• Participated in multiple technical competitions and collaborative events
• Participated in several debates, and other literary activities

Certifications

Programming for Everybody (Getting Started with Python) — University of Michigan

Python Data Structures — University of Michigan

AI Tools and ChatGPT Workshop — Be10x

Languages

English — Fluent
Bengali — Native
Hindi — Conversational